



Homework 7

Deadline: 30 November 2025, 23:30.

All solutions must be in a single PDF file and uploaded to the LMS portal.

1. (0.25 point) Derive the expression for $\frac{\partial V}{\partial S}$ for a geometric Asian call option within the modified Black-Scholes- Merton model with estimated volatility function from GARCH(1,1).
2. (0.25 point) Derive the expression for $\frac{\partial^2 V}{\partial S^2}$ for a geometric Asian call option within the modified Black-Scholes- Merton model with estimated volatility function from GARCH(1,1).
3. (0.5 point) Consider the instantaneous forward rate $f(t, T) = \lim_{\delta \rightarrow 0} F(t, T, T + \delta)$. Find the relationship between the rates $f(t, T_1)$ and $f(t, T_2)$, where $T_1 < T_2$, in the Hull-White model¹ with parameters $\theta(t) = \theta$ and $\sigma(t) = \sigma$.

¹See Lecture 7 Section 2.